Mapping microbiome induced host transcriptomic interactions in colorectal cancer

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Table of contents:

Acknowledgements 3

Abstract 4

Introduction 5-7

Aims and objectives 8

Methodology 9-13

Results 14-21

Discussion and conclusion 22-25

Bibliography 26-29

Acknowledgements

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Abstract

Fusobacterium nucleatum (*Fn*), a gut-associated microbe implicated in colorectal cancer (CRC), is known to drive tumour progression, inflammation and chemoresistance. However, how *Fn* influences tumour biology across distinct molecular subtypes of CRC remains poorly understood. This study investigates subtype-specific transcriptomic responses to *Fnn (F. nucleatum* subsp. *nucleatum)* infection using HT-29 and MDST8 cell lines, representative of CMS1 and CMS4 CRC subtypes, respectively.

Cells were infected with *Fnn* for 4 hours under optimised conditions that preserved viability and elicited a robust inflammatory response. RNA sequencing and differential expression analysis identified 18 significantly altered genes in HT-29 and 4 in MDST8, with *DUSP1* being the only gene downregulated in both models. Enrichment analysis in CMS1-like HT-29 cells revealed strong enrichment of pathways involved in protein synthesis and translation including ribosome biogenesis and mitochondrial translation. In contrast, CMS4-like MDST8 cells exhibited enrichment in transcription factor and microRNA target gene sets, suggesting subtype-specific regulation at transcriptional and post-transcriptional levels.

Network and exploratory pathway analyses further supported distinct functional responses between subtypes. The consistent downregulation of *DUSP1* across both models suggests a shared microbial stress-response mechanism involving MAPK pathway regulation.

These findings demonstrate that *Fnn* elicits both divergent and convergent transcriptional programs across CRC subtypes. This work underscores the value of subtype-stratified models in microbiome-cancer research and may inform future therapeutic strategies tailored to tumour molecular context.

Introduction

Colorectal cancer (CRC) is the third most diagnosed cancer globally and the second leading cause of cancer-related mortality, responsible for nearly 1 million deaths annually worldwide (1). In Ireland, CRC represents the second most common cancer diagnosis across genders (2). Despite improvements in screening and therapy, CRC outcomes vary significantly between patients due to the underlying molecular and cellular heterogeneity of CRC tumours (3). The Colorectal Cancer Subtyping Consortium (CRCSC) therefore established distinct Consensus Molecular Subtypes (CMS) of CRC tumours based on gene expression profiles, demonstrating that CRC is not a homogeneous disease but comprises multiple subgroups with unique biology (4). This molecular heterogeneity shapes tumour behaviour and contributes to differences in treatment outcomes. Notably, CMS1 tumours (accounting for ~14% of CRC) are typically microsatellite instability-high (MSI-H) with strong immune infiltration, whereas CMS4 (~23%) exhibits invasive mesenchymal characteristics along with stromal activation and enhanced angiogenesis (5). CMS4 cancers are associated with the poorest prognosis among all subtypes, likely due to their chemoresistant profile, while CMS1 shows the most favourable outcomes (5,6). These differences underscore the need for stratified approaches to CRC research and treatment.

Growing evidence has implicated the gut microbiome, and particularly the gram-negative anaerobe *Fusobacterium nucleatum (Fn)*, in CRC pathogenesis and patient outcomes. *Fn* is commonly enriched within CRC tissues and adenomas, while rarely present in the gastrointestinal tract of healthy individuals (7). It has been implicated in tumorigenesis, metastasis and treatment resistance in CRC (3,7–9). Specifically, *Fn* is associated with late-onset CRC, where higher abundance of the bacteria is associated with poorer overall survival (8,10,11). Mechanistically, *Fn* promotes tumour progression through several pathways: it activates pro-inflammatory NF-κB signalling via Toll-like receptor 4 (TLR4)/MYD88 interaction (12), stimulates Wnt/ β-catenin signalling through FadA-mediated E-cadherin binding (13), and contributes to chemoresistance by suppressing apoptosis and enhancing autophagy (9). These mechanisms are summarised in **Figure 1**, which illustrates key *Fn*-mediated signalling pathways that contribute to tumour proliferation.

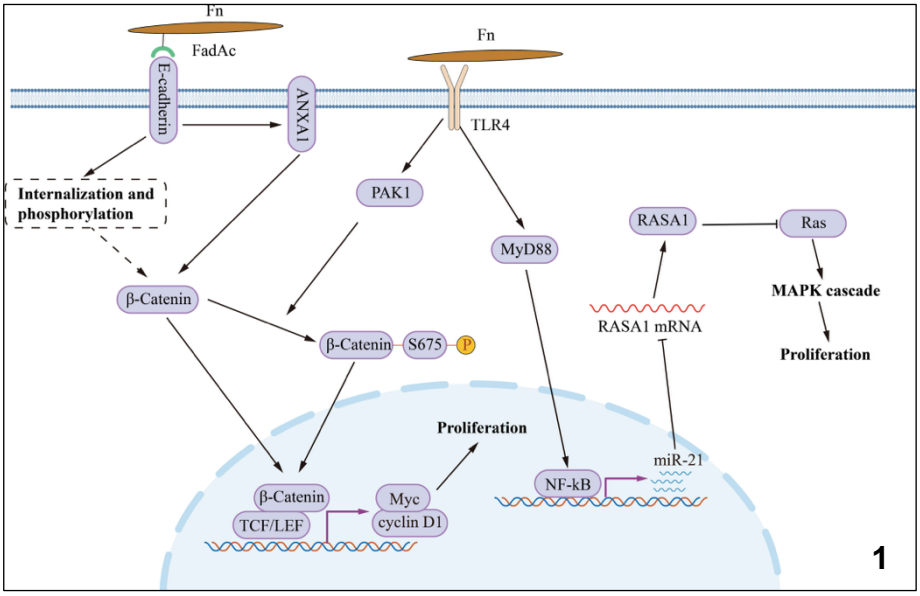


Figure 1. *Fn*-mediated activation of oncogenic signalling pathways in CRC.
Adapted from Ou et al (2022).
Fn promotes colorectal cancer cell proliferation through activation of Wnt/ β-catenin and TLR4/MyD88-NF-κB signalling. These pathways upregulate pro-proliferative genes (e.g. Myc, cyclin D1) and suppress tumour suppressor activity via miR-21, facilitating tumour progression (14).

Fn also modulates the tumour microenvironment by suppressing cytotoxic T-cell infiltration and promoting the accumulation of myeloid-derived suppressor cells, thereby facilitating immune evasion (15,16). More recently, metabolic interactions between *Fn* and host cells have been a subject of interest (15,17) . *Fn*-derived formate and short-chain fatty acids can activate the aryl hydrocarbon receptor (AhR), a transcription factor known to regulate cancer stem cell phenotypes, detoxification pathways, and immune responses (17–19).

Despite growing evidence of *Fn*'s pathogenic role in CRC, most studies have used single-cell line models and failed to account for tumour subtype heterogeneity. Given that CMS1 and CMS4 have distinct immune, stromal and transcriptional landscapes, it is plausible that *Fn* exerts differential effects across subtypes. For instance, CMS1 reactive tumours may elicit stronger immune responses to bacterial infection, while CMS4 tumours, which are more immunosuppressive and chemoresistant, may engage alternative survival pathways (6,20). These subtype-specific differences suggest that host-microbiome interactions, including responses to *Fn*, may be context-dependent and contribute to divergent outcomes.

Very limited studies have directly examined *Fn*'s subtype-specific effects at the transcriptomic level. This gap in the literature limits our understanding of host-microbe interactions in CRC and undermines efforts to develop subtype-informed therapeutic strategies. Stratifying *Fn*'s impact according to CMS subtype could provide mechanistic insight into differential responses to bacterial colonisation and uncover novel therapeutic vulnerabilities.

This study aims to address this gap by investigating *Fnn*-induced transcriptional responses in vitro models representing CMS1 and CMS4 subtypes. HT-29 and MDST8 cell lines were selected as representative models of CMS1 and CMS4, respectively, based on their classification in colorectal cancer literature (6,21). It is important to note that HT-29 has shown conflicting evidence supporting CMS1 and CMS3 classification across studies (21,22). This analysis focuses on identifying subtype-specific transcriptional alterations, as well as any shared responses to *Fnn* infection.

By characterising *Fn*'s influence on subtype-specific transcriptional programs, this study seeks to advance understanding of how *Fn* interacts with distinct tumour molecular contexts. These findings may offer insights into how *Fn* modulates tumour biology across different CRC subtypes and support further investigations into CMS-informed therapeutic strategies in CRC.

Aims and Objectives

Hypothesis:
Fusobacterium subsp*. nucleatum* (*Fnn*) induces distinct transcriptomic responses in CMS1 and CMS4 colorectal cancer cell models due to their divergent molecular characteristics that may contribute to CRC pathogenesis.

Objective:
The overall objective of this study was to investigate how *Fnn* modulates transcriptional responses in colorectal cancer cell lines representing distinct Consensus Molecular Subtype (CMS).

The specific aims were to:

Aim 1: Evaluate the transcriptomic changes induced by *Fnn* infection in HT-29 (CMS1-like) and MDST8 (CMS4-like) colorectal cancer cell lines.

Aim 2: Identify shared and subtype-specific differentially expressed genes in response to *Fnn.*

Aim 3: Assess functional relationships among differentially expressed genes using protein-protein interaction network analysis (STRING).

Aim 4: Investigate biological pathways and gene programs affected by *Fnn* by using Gene Set Enrichment Analysis (GSEA) and exploratory KEGG pathway analysis.

Methodology

Cell culture

HT-29 and MDST8 human CRC cell lines were used to investigate the transcriptomic effects of *Fusobacterium nucleatum* subsp. *nucleatum (Fnn)* infection. Both cell lines were cultured in Dulbecco's Modified Eagle Medium (DMEM; high glucose, GlutaMAX; Thermofisher), supplemented with 10% foetal bovine serum (FBS; Gibco, Invitrogen) and 1% penicillin-streptomycin (Gibco, Invitrogen). Cells were maintained in a humidified incubator at 37°C with 5% CO_2 and grown to 70% confluency before experimental treatment.

Bacteria culture and infection

Fusobacterium nucleatum subsp. *nucleatum* (strain VPI 4355 [1612A]) was obtained from ATCC (ATCC 25586 ™). It was cultured in Oxoid Fastidious Anaerobe Broth (Thermo Scientific) and on TSA plates supplemented with 10% sheep's blood (Fannin), and maintained in an anaerobic chamber (90% N2) at 37 °C. Primary cultures were grown for 24-48 hours under anaerobic conditions. Bacterial concentrations were determined using colony-forming unit (CFU) enumeration and OD_{600} measurements to establish accurate multiplicity of infection (MOI) ratios for cell treatments. *Fnn* was resuspended in PBS and added to cultured cells at four different MOIs of 1:1000, 1:100, 1:50, and 1:10. An overview of the experimental workflow leading to RNA-sequencing is shown in **Figure 2**.

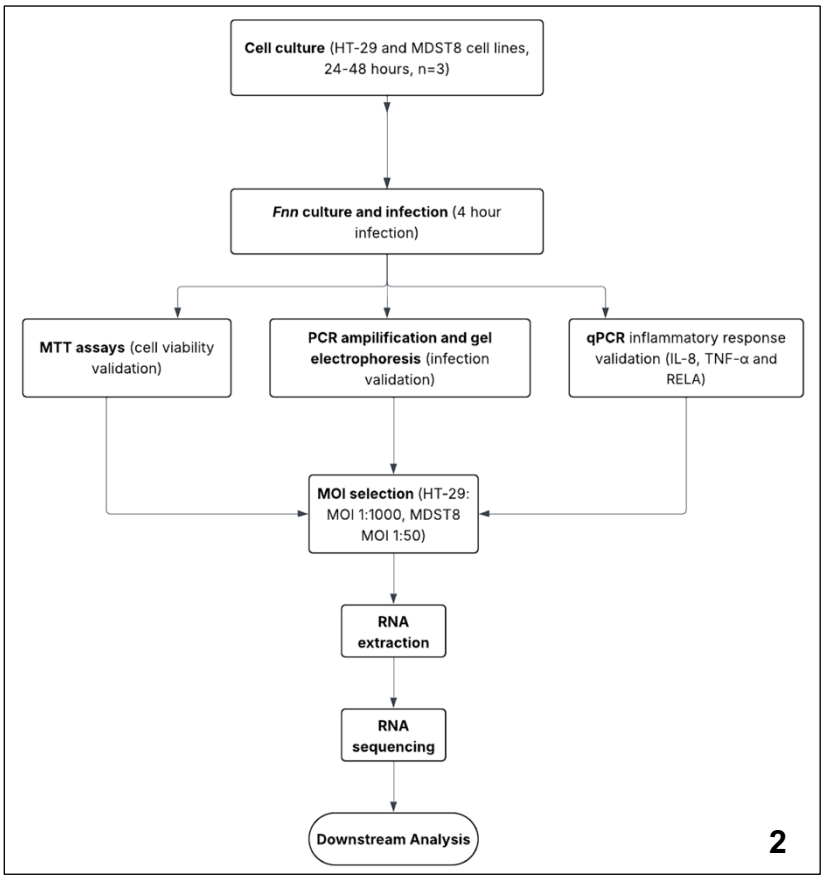


Figure 2. Experimental workflow summarising key steps in sample preparation, validation, and RNA sequencing.
HT-29 and MDST8 cells were infected with F*nn* and assessed for cell viability (MTT), infection (PCR), and inflammatory response (qPCR) validation before RNA was extracted and sequenced.

Cell viability was assessed using MTT assays at 4 hours post-infection across all MOI ratios. Both cell lines maintained >95% viability across the tested MOI range (**Figure 3**).

Successful *Fnn* infection was confirmed by PCR amplification of the fadA gene (232 bp amplicon), visualised via gel electrophoresis (**Figure 4**). Quantitative PCR analysis was used to assess inflammatory marker response (*IL-8*, *TNF-α*, and *RELA*). *IL-8* expression was significantly upregulated in infected cells relative to PBS-treated controls (HT-29: 2-fold increase at MOI 1:1000; MDST8: 1.3-fold increase at MOI 1:50) (**Figure 5**).

Based on these optimisation results, the selected conditions for RNA sequencing (conducted by Novogene) were: HT29 cells treated with *Fnn* (MOI 1:1000) and matched PBS controls, and MDST8 cells treated with *Fnn* (MOI 1:50) and the corresponding controls. All samples passed quality control checks prior to sequencing.

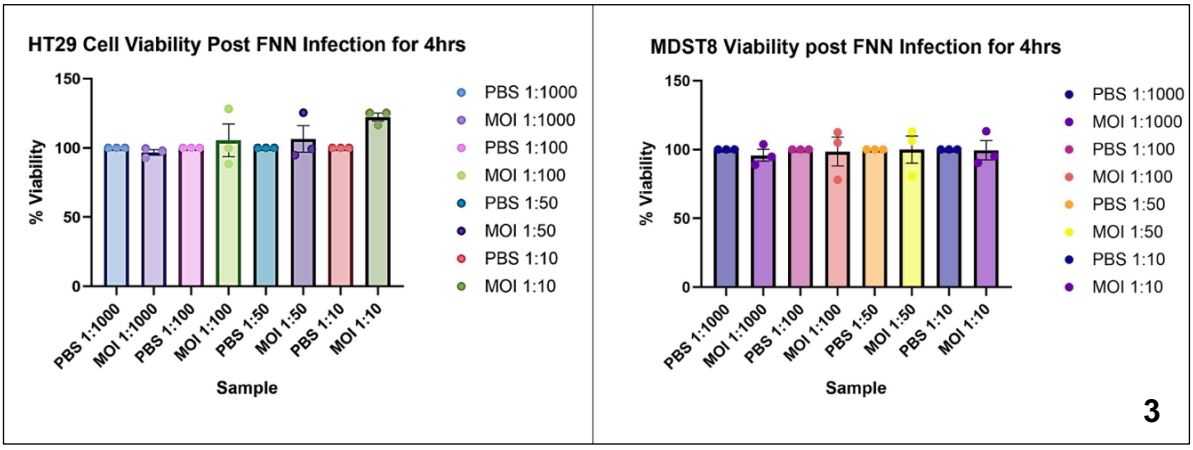


Figure 3. Cell viability assessment demonstrates high cell viability maintained across MOI ratios for both cell lines.
MTT cell viability assays showing percent viability of HT-29 (left panel) and MDST8 (right panel) cells at 4 hours post-*Fnn* infection across multiple MOI ratios (1:10 to 1:1000). Both cell lines maintained >95% viability across all tested MOI ratios compared to PBS controls, demonstrating no significant change in cell viability. Data presented as mean ± SEM (n=3 biological replicates). Statistical significance assessed by t-test (*p<0.05).

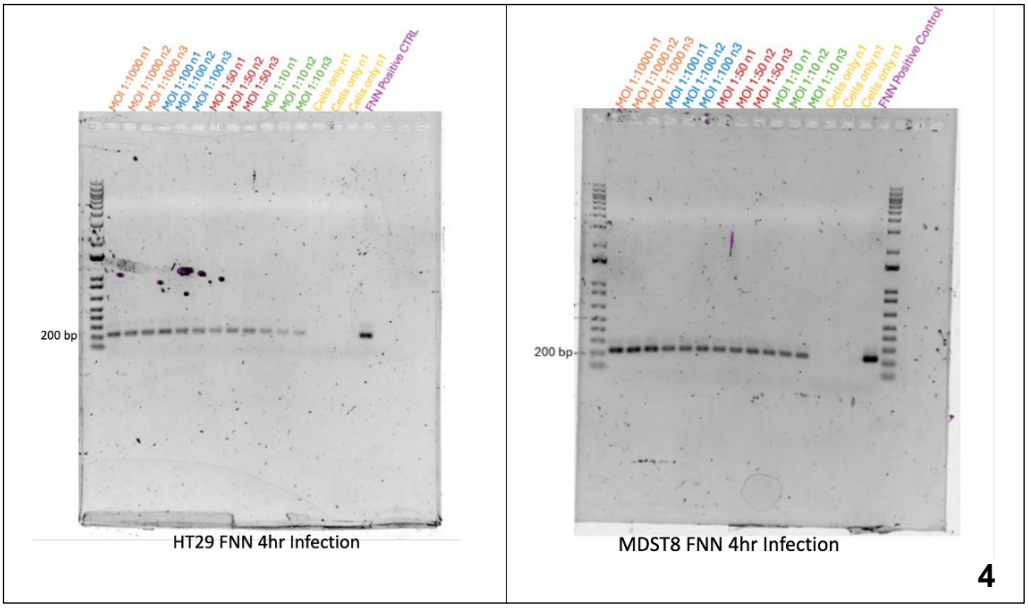


Figure 4. Gel electrophoresis confirmation of *Fnn* infection in both cell lines. Agarose gel electrophoresis showing PCR amplification of *Fnn* fadA gene (232bp) in HT-29 (left panel) and MDST8 (right panel) cells at 4 hours post-infection. Successful bacterial infection is confirmed by the presence of the expected 232bp band in infected samples across multiple MOI ratios. Ladder shows 200bp and 1bp markers.

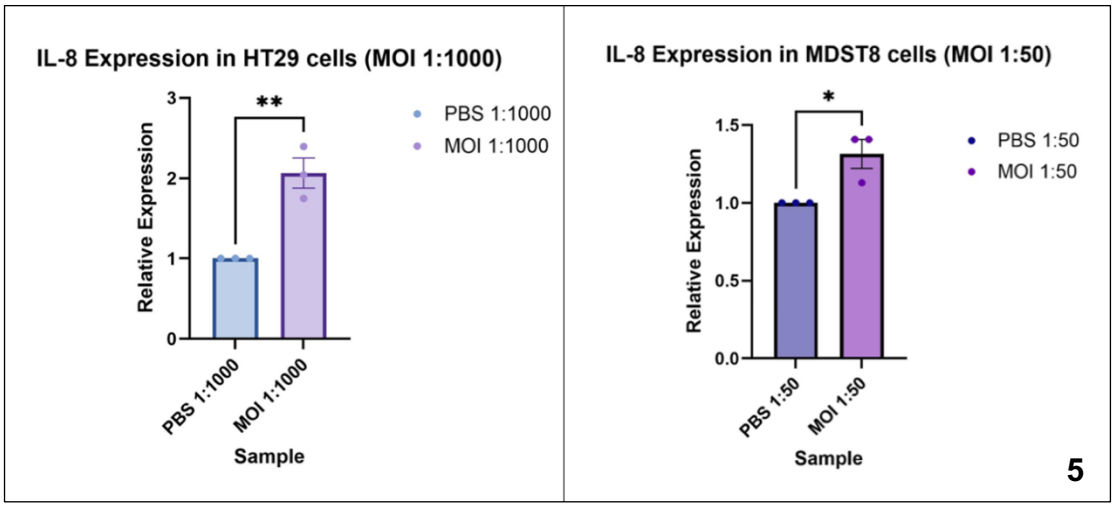


Figure 5. Quantitative PCR validation of inflammatory response at selected MOI ratios.
IL-8 gene expression analysis by qPCR showing inflammatory response in HT-29 cells at MOI 1:1000 (left panel) and MDST8 cells at MOI 1:50 (right panel) compared to PBS controls at 4 hours post-*Fnn* infection. Both cell lines demonstrated significant upregulation of *IL-8* expression (HT-29: 2-fold increase; MDST8 1.3-fold increase) confirming robust inflammatory response at the selected MOI ratios used for RNA sequencing. Data presented as mean ± SEM (n=3 biological replicates). Statistical significance assessed by t-test (*p<0.05, **p<0.01).

RNA sequencing, data preprocessing and quality assessment

Raw sequencing reads were processed using the nf-core/rnaseq pipeline (23), a standardised and reproducible workflow designed to ensure consistent data processing and minimise analytical bias. This pipeline includes quality control, read trimming, genome alignment, and gene expression quantification. FastQC reports confirmed that all samples met quality thresholds, indicating high sequencing quality suitable for downstream analysis. Reads were aligned to the human reference genome GRCh38/hg38.

Principal component analysis (PCA) and sample clustering

Principal component analysis (PCA) was performed to assess sample clustering, validate transcriptional separation between HT-29 and MDST8 cell lines and evaluate technical variability across triplicates. PCA was conducted on variance-stabilising transformed (VST) data using DESeq2 and visualised with ggplot2 (24). As shown in **Figure 6** (Results), PCA confirmed clear separation between cell lines, although some variability among biological triplicates was observed.

Differential expression analysis

Differential expression analysis was conducted using DESeq2 version 1.44.0 (25) and EdgeR version 4.4.2 (26), both implemented through Bioconductor version 3.19 (27) in R version 4.4.1. DESeq2 and EdgeR were applied in parallel to provide cross-validation, as each employs a different statistical framework (negative binomial vs. quasi likelihood). Analyses were performed separately for each cell line to identify subtype-specific responses to *Fnn* infection.

DESeq2: Raw count matrices were normalised using the median-of-ratios method to account for library size and composition. Differential gene expression was assessed using DESeq2's negative binomial generalised linear model and Wald test, comparing infected and control samples within each cell line.

edgeR: Differential analysis was performed using the quasi-likelihood (QL) pipeline. Raw counts were normalised using TMM (trimmed mean of M-values), followed by QL F-tests via the glmQLFTest function to identify differential expressed genes in each cell line.

Statistical Thresholds: For primary analysis, differentially expressed genes were defined using an adjusted p-value<0.05 and a log2 fold change<0.58 (equivalent to 1.5-fold change). Gene annotation was performed using Ensembl gene identifier converted to gene symbols for biological interpretation.

Visualisation: DE results were visualised using volcano plots and heatmaps. Volcano plots were generated using ggplot2 and ggrepel (28) packages to display log2fold change versus -log10(adjusted p-value). Heatmaps were created with the pheatmap(29) package, using Z-score normalised, log2-transformed expression values with hierarchical clustering.

Method selection: While both DESeq2 and EdgeR analyses were performed to validate gene expression trends, DESeq2 results were prioritised for downstream enrichment analyses.

Network analysis

STRING protein-protein interaction (PPI) analysis was performed on differentially expressed genes (adjusted p-value<0.05) to identify any predicted functional relationships and coordinated regulatory networks between significantly differentially expressed genes. PPI enrichment p-values were assessed to determine the functional relevance of the interaction networks identified for HT-29 and MDST8.

Gene Set Enrichment Analysis and pathway analysis

Gene Set Enrichment Analysis (GSEA) was performed using GSEA software version 4.4.0 from the Broad Institute (30,31). Analyses were run against multiple collections from the Molecular Signatures Database (MSigDB) (32), collections including Hallmark (H), Canonical Pathway (CP), Regulatory Target (C3), Computational (C4), Oncogenic Signatures (C6) and Immunologic Signature (C7) gene sets.

Standard parameters included 1,000 permutations for significance testing and a minimum gene set size of 15 genes. Although an FDR (False Discovery Rate) threshold of <0.27 was set to allow exploratory interpretation across both cell lines, in practice enriched gene sets in HT-29 cells showed highly significant values (FDR<0.0075) and were interpreted directly. In contrast, in MDST8 cells, several microRNA target gene sets showed FDR values between 0.25-0.268. While these higher FDR values in MDST8 were not used for direct biological interpretation, they provided supporting rationale for the observed negative enrichment of transcription factor target gene sets. Thus, the relaxed FDR threshold was applied selectively to ensure meaningful enrichment results were captured while maintaining biological interpretability.

Pathway enrichment analysis

KEGG pathway analysis was conducted as an exploratory approach using differentially expressed genes (DEGs) that met a nominal p-value threshold of <0.05. This less stringent cutoff was selected due to the limited number of DEGs identified at adjusted p-value<0.05, which prevented significant enrichment at conventional thresholds. Further limiting statistical power, gene ID mapping issues prevented 22.22% of HT-29 genes and 25% of MDST8 DEGs from being annotated with KEGG Entrez identifiers. Enrichment analysis was conducted using the clusterProfiler R package (33).

Results

Principal component analysis (PCA) demonstrates clear cell line separation but minimal treatment effects

Principal component analysis revealed clear transcriptional separation between HT-29 and MDST8 cell lines along PC1 (91.4% variance), confirming distinct molecular profiles consistent with their respective molecular subtypes (CMS1 & CMS4 respectively) (**Figure 6**). However, triplicates showed more variability than expected, suggesting technical variation, biological variation or experimental noise between samples. Additionally, minimal treatment versus control clustering within each cell line suggested that *Fnn* effects are subtle. These observations informed expectations for differential expression analysis and justified conducting separate analyses for each cell line to identify subtype-specific responses to *Fnn.*

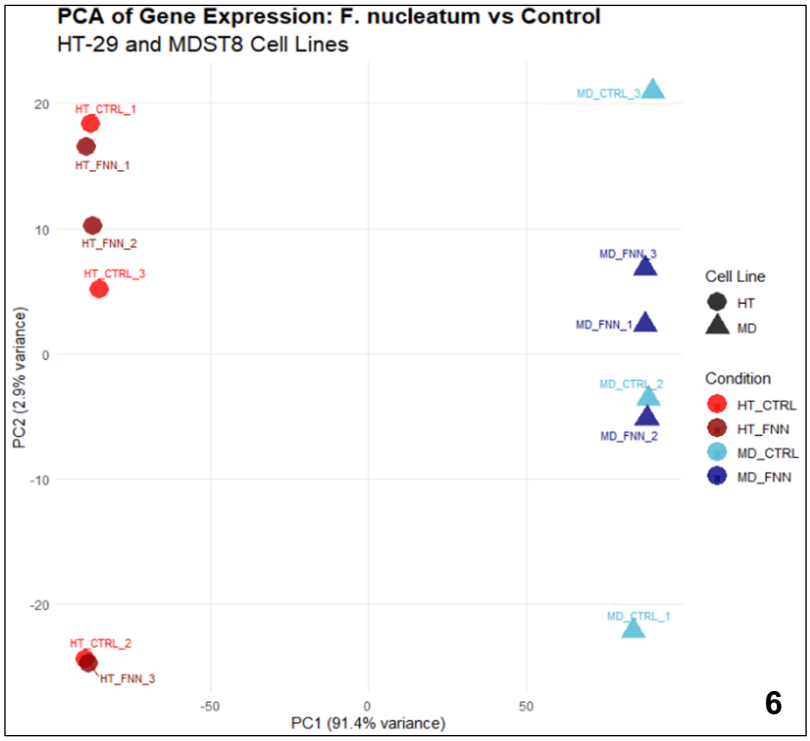


Figure 6. Principal component analysis demonstrates clear molecular subtype separation but minimal treatment vs control clustering.
PCA plot showing transcriptional profiles of HT-29 (circles) and MDST8 (triangles) cell lines under control (CTRL) and *Fnn* infection (FNN) conditions. PC1 accounts for 91.4% of variance and clearly separates the two cell lines, confirming distinct molecular profiles consistent with CMS1 (HT-29) and CMS4 (MDST8) colorectal cancer subtypes. PC2 accounts for 2.9% of variance. Triplicates show more variability than expected, suggesting technical variability, biological variation or experimental noise between samples. Minimal treatment versus control clustering within each cell line indicates *Fnn* effects are subtle, which informed expectations for differential expression analysis.

Differential expression analysis reveals limited but distinct cell line-specific responses to *Fnn* infection

To identify genes responsive to *Fnn* infection, differential expression analysis was performed separately for each cell line with a log2fold change threshold of 0.58 (1.5-fold). Using stringent criteria (adjusted p-value<0.05), HT-29 showed a broader transcriptional response with 18 significantly differentially expressed genes, while MDST8 demonstrated a more limited response with only 4 genes meeting the criteria (**Table 1**). Notably, only one gene (*DUSP1*) overlapped between the two cell lines under these stringent criteria (**Figure 7A**).

The expression patterns and magnitude of fold changes among differentially expressed genes were visualised using heatmaps and volcano plots for each cell line (**Figure 7B, 7C**).

Cell Line	Gene	log2FC	p-value	adj. p-value	Direction
HT-29	*AHRR*	1.6	2.94E-06	0.004	**Upregulated**
HT-29	*ATP8B4*	3.09	2.62E-05	0.028	**Upregulated**
HT-29	*FAM154B*	5.87	4.58E-05	0.046	**Upregulated**
HT-29	*TRPV1*	4.69	9.70E-07	0.001	**Upregulated**
HT-29	*NEAT1*	1.1	1.53E-07	0	**Upregulated**
HT-29	*THBS1*	1.16	5.54E-08	0	**Upregulated**
HT-29	*ENPEP*	2.06	2.44E-05	0.028	**Upregulated**
HT-29	*CYP1A1*	3.4	7.30E-34	0	**Upregulated**
HT-29	*GPR110*	0.87	2.20E-07	0	**Upregulated**
HT-29	*ALDH1A3*	1.48	1.35E-07	0	**Upregulated**
HT-29	*RP11-66B24.4*	1.09	2.45E-07	0	**Upregulated**
HT-29	*RP11-504P24.6*	9.11	4.30E-13	0	**Upregulated**
HT-29	*DUSP1*	-0.94	2.70E-08	0	**Downregulated**
HT-29	*ID1*	-0.72	1.38E-06	0.002	**Downregulated**
HT-29	*RP11-504P24.4*	-2.73	2.76E-07	0	**Downregulated**
HT-29	*SLC5A3*	-12.84	8.01E-27	0	**Downregulated**
HT-29	*AL162407.1*	-8.38	7.32E-11	0	**Downregulated**
HT-29	*RP11-1143G9.4*	-1.63	5.10E-05	0.049	**Downregulated**
MDST8	*TXNIP*	1.13	4.21E-06	0.026	**Upregulated**
MDST8	*DUSP1*	-0.98	6.70E-07	0.005	**Downregulated**
MDST8	*EGR1*	-1.47	3.68E-12	0	**Downregulated**
MDST8	*FOSB*	-1.2	1.74E-08	0	**Downregulated**

Table 1: Significantly differentially expressed genes at adjusted p-value <0.05 (DESeq2).
List of genes meeting stringent statistical criteria (adjusted p-value <0.05, log2 fold change <0.58) showing reduced gene sets that maintain the pattern of distinct

responses between cell lines. *DUSP1* is the only gene consistently identified (downregulated) in both cell lines under stringent criteria.

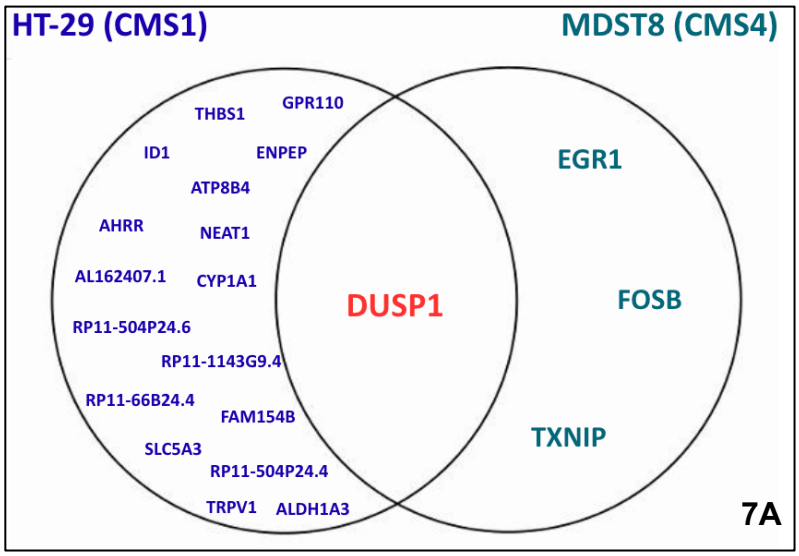
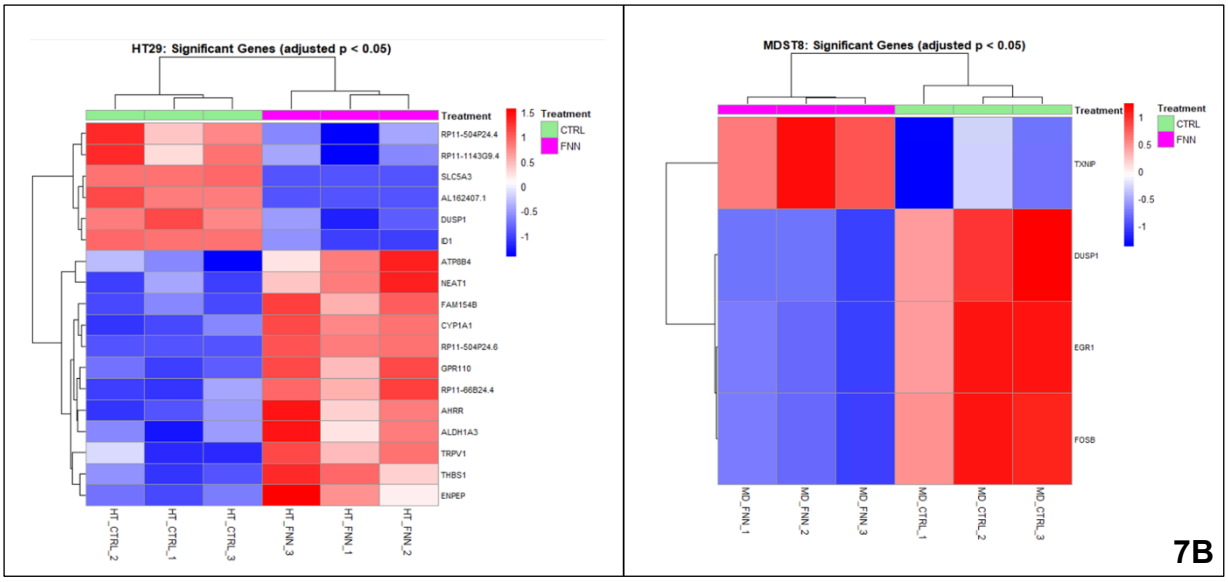


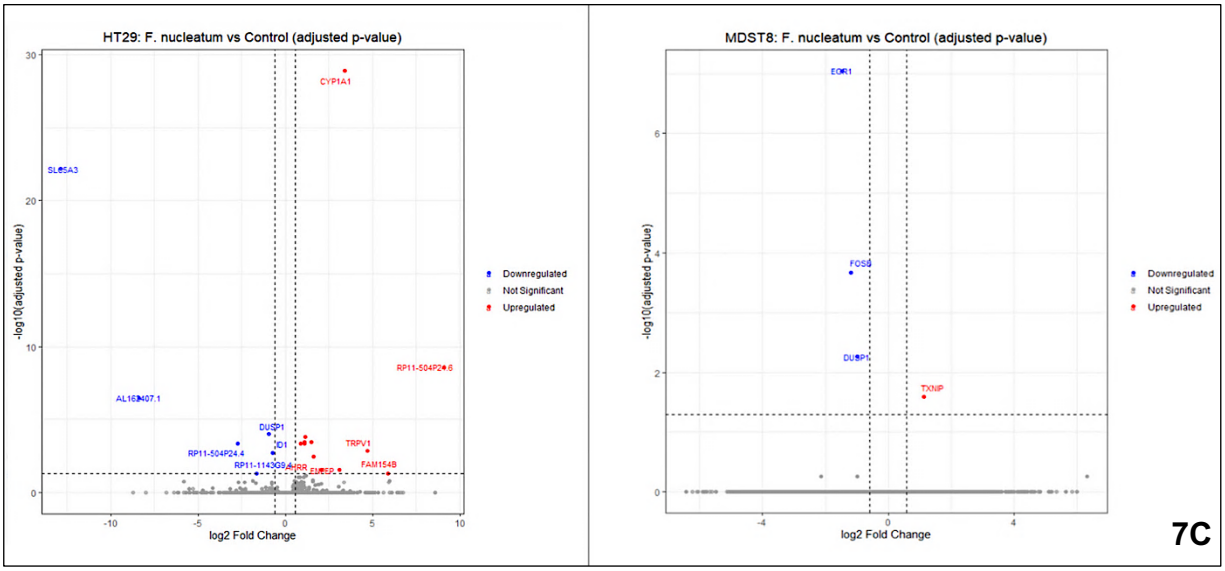
Figure 7. Differential expression analysis reveals distinct cell line-specific responses to *Fnn* infection

(A): Venn diagrams demonstrating distinct subtype-specific transcriptional responses to *Fnn* infection. Left panel shows overlap of differentially expressed genes at adjusted p-value <0.05, with HT-29 showing 18 genes and MDST8 showing 4 genes, with only *DUSP1* overlapping between cell lines.

(B): Heatmaps displaying significantly differentially expressed genes (DESeq2, p<0.05) for HT-29 (left panel, 18 genes) and MDST8 (right panel, 4 genes) cell lines following *Fnn* infection. Expression values are Z-score normalised and clustered by similarity. Only 1 gene (*DUSP1*), overlaps between the cell lines. Red indicates higher expression, while blue indicates lower expression for each gene across samples (n=3 biological replicates per condition). Clear treatment-based clustering is observed within each cell line, demonstrating the differential expression patterns.

(C) Volcano plots showing statistical significance versus fold change for genes meeting stringent criteria (adjusted p-value<0.05) in HT-29 (left) and MDST8 (right) cell lines following *Fnn* infection. X-axis represents log2 fold change, y-axis represents -log10(adjusted p-value). Vertical dashed lines indicate log2 fold change thresholds (±0.58), horizontal dashed line indicates adjusted p-value threshold (0.05). Red dots represent upregulated genes, blue dots represent downregulated genes, grey dots represent non-significant genes. Key significant genes are labelled, demonstrating the broader transcriptional response in HT-29 (18 genes) compared to more focused response in MDST8 (4 genes).





STRING network analysis identifies functional gene clusters

To understand the functional relationships among the biologically significant genes, protein-protein interaction network analysis was performed using STRING database. Network analysis revealed highly significant protein-protein interactions between differentially expressed genes in each cell line, suggesting coordinated regulatory responses to *Fnn* infection, rather than random gene expression changes.

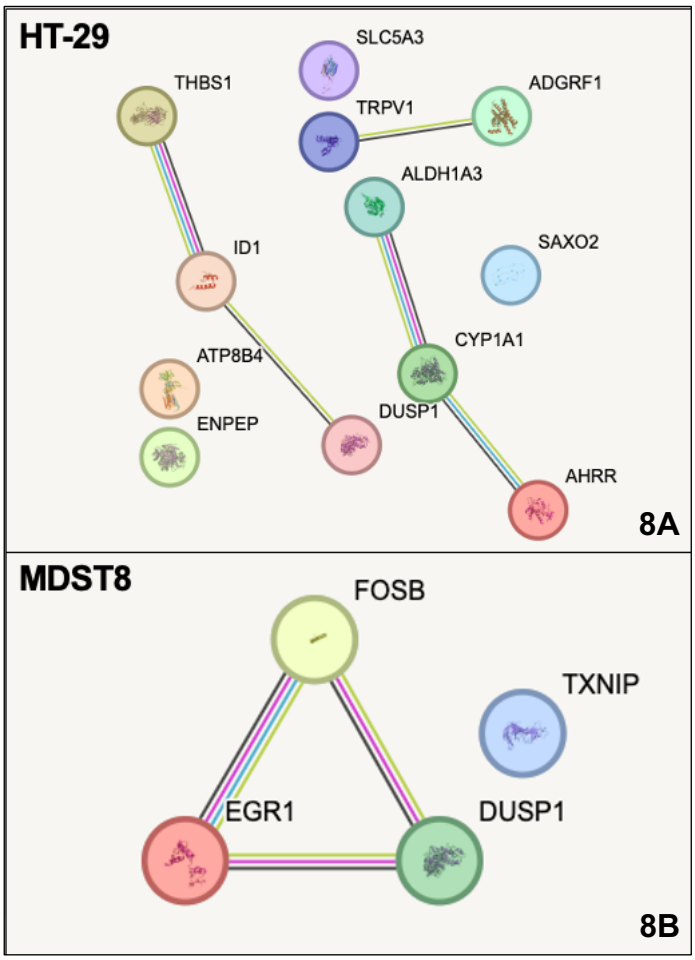


Figure 8: STRING protein-protein interaction networks of differentially expressed genes reveal coordinated functional clusters.
(A): STRING network analysis of significantly differentially expressed genes (adjusted p-value <0.05) in HT-29 (upper panel) cells following *Fnn* infection. Each node represents a differentially expressed gene, and edges indicate known protein-protein interactions from the STRING database. Line thickness represents interaction confidence scores. HT-29 shows genes including *CYP1A1*, *AHRR* and *DUSP1*, forming functional clusters. This network demonstrates significant pathway enrichment (p = 0.000296), suggesting non-random co-regulation.
(B): STRING network analysis of differentially expressed genes (adjusted p-value<0.05) in MDST8 cells post-*Fnn* infection. A tight regulatory cluster is observed, centred on *FOSB*, *EGR1* and *DUSP1* . This network also shows significant interaction enrichment (p = 0.000377), indicating coordinated gene regulation in CMS4-like cells.

Gene Set Enrichment Analysis (GSEA) reveals distinct pathway responses between CMS subtypes

To understand the biological significance of the differentially expressed genes, Gene Set Enrichment Analysis (GSEA), was performed using the Human Molecular Signatures Database (MSigDB) collections. GSEA analysis revealed markedly different pathway enrichment patterns between the two cell lines, suggesting that *Fnn* infection triggers distinct molecular programs in CMS1 versus CMS4 colorectal cancer subtypes (**Figure 9A** and **9B**).

In HT-29 cells (CMS1 model), *Fnn* infection primarily upregulated pathways related to protein synthesis and cellular metabolism. The most significantly enriched pathways include ribosome biogenesis (FDR<0.001), eukaryotic translation elongation (FDR<0.001), and SRP-dependent cotranslational protein targeting (FDR<0.001) gene sets. These findings suggested that CMS1 subtype responds to bacterial infection by enhancing protein synthesis machinery and metabolic processes.

In contrast, MDST8 cells (CMS4 model) demonstrated a distinct enrichment pattern focused on regulatory mechanisms. The most significantly enriched pathways included transcription factor target gene sets (*TFAM* and *GTF3A*, FDR<0.1), and microRNA target gene sets (MIR548, MIR3065, MIR501). This pattern indicated that the CMS4 subtype preferentially activates post-transcriptional and epigenetic regulatory mechanisms in response to *Fnn* infection.

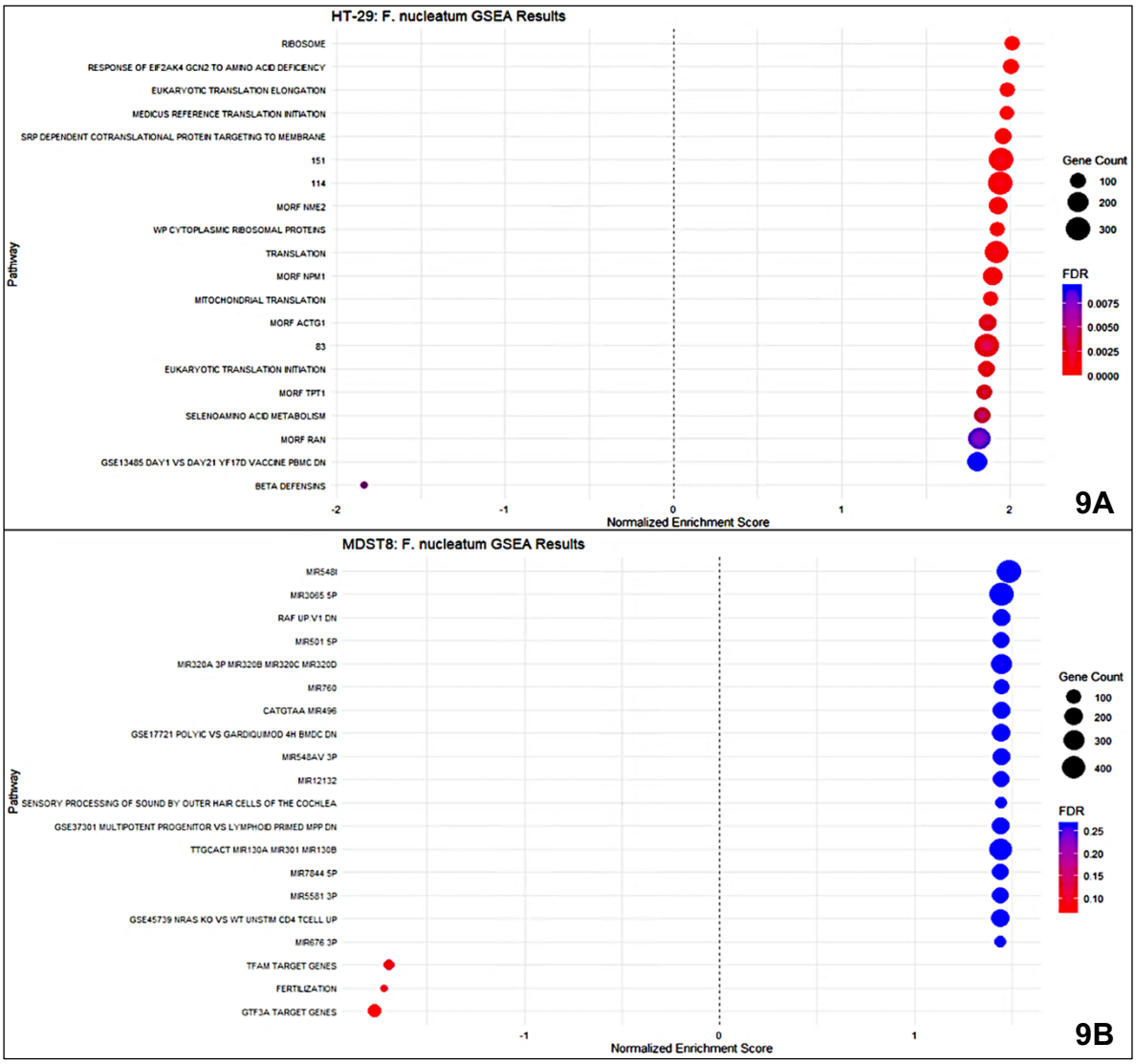


Figure 9. Gene Set Enrichment Analysis (GSEA) reveals distinct pathway responses to *Fnn* infection between CMS subtypes.
(A): Bubble plot showing significantly enriched pathways in HT-29 (upper panel) and MDST8 (lower panel) cell lines following *Fnn* treatment (n=3 biological replicates per condition). X-axis represents normalised enrichment score (NES) with positive values indicating pathway upregulation and negative values indicating downregulation. Bubble size represents gene set size, bubble colour represents FDR q-value (red= most significant, blue= least significant). HT-29 cells show strong enrichment in protein synthesis and metabolic pathways, including ribosome biogenesis, translation elongation, and mitochondrial translation.
(B): Bubble plot displaying significantly enriched pathways in MDST8 cells after *Fnn* infection (n=3 biological replicates per condition). Enriched gene sets primarily include transcription factor targets (e.g. TFAM, GTF3A) and less significantly enriched microRNA target gene sets (e.g. MIR548, MIR3065), indicating transcriptional and post-transcriptional regulatory activity. Bubble size and colour represent gene count and FDR significance, respectively. Vertical dashed line at NES=0 indicates no enrichment.

Exploratory KEGG analysis supports evidence of distinct functional responses

To complement GSEA findings, KEGG pathway analysis was performed on differentially expressed genes (DEGs) that met a nominal significance threshold of $p<0.05$. This exploratory analysis was necessary due to the limited number of DEGs meeting adjusted p-value thresholds and the occurrence of gene ID mapping failures (22.22% of HT-29 genes and 25% of MDST8 genes failed to map to KEGG Entrez identifiers).

KEGG analysis revealed distinct pathway enrichment patterns between the two cell lines. HT-29 showed enrichment of stress-related pathways including MAPK signalling, chemical carcinogenesis involving reactive oxygen species and TNF signalling. In contrast, MDST8 demonstrated enrichment of cell communication pathways including cAMP signalling, axon guidance and IL-17 signalling. These exploratory findings support GSEA results showing distinct functional responses to *Fnn* between CMS subtypes.

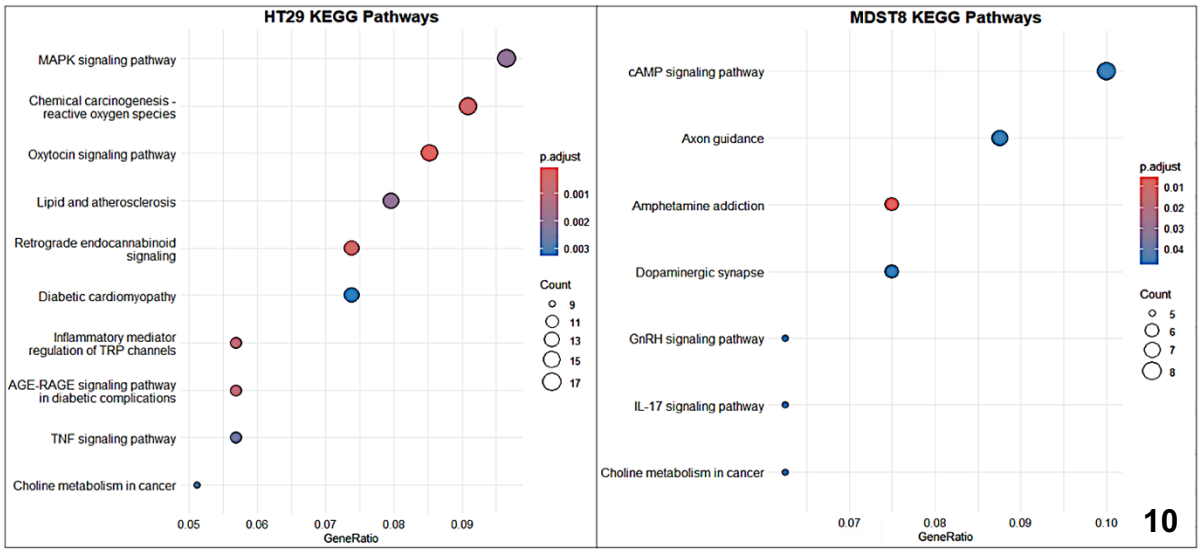


Figure 10: Exploratory KEGG pathway analysis supports findings of distinct functional enrichment patterns between CMS subtypes.
Bubble plots showing enriched KEGG pathways (DEGs with p-value <0.05) in HT-29 (left panel) and MDST8 (right panel) cell lines following *Fnn* infection. X-axis represents gene ratio (proportion of differentially expressed genes in each pathway), Y-axis lists enriched pathways. Bubble size indicates gene count per pathway, bubble colour represents adjusted p-value significance (red= most significant, blue= least significant). HT-29 demonstrates enrichment of stress and inflammatory pathways including MAPK signalling, reactive oxygen species pathways, TNF signalling and inflammatory mediator regulation. MDST8 shows enrichment of cell communication pathways including cAMP signalling, axon guidance, dopaminergic synapse and IL-17 signalling. Both cell lines show choline metabolism in cancer enrichment. Analysis was performed using DEGs meeting a p-value <0.05 threshold due to limited gene counts and gene ID mapping constraints (22.22% of HT-29 genes and 25% of MDST8 genes failed to map to KEGG identifiers).

Discussion

This study aimed to evaluate whether *Fusobacterium nucleatum* subsp. *nucleatum* (*Fnn*), a gut-colonising bacterium linked to colorectal cancer (CRC) progression and chemoresistance (9,34), induces distinct transcriptomic responses in different CRC molecular subtypes. HT-29 and MDST8 cell lines, representing Consensus Molecular Subtypes CMS1 and CMS4 respectively (21), were infected with *Fnn* and subjected to transcriptomic profiling using RNA sequencing. Differential expression (DE) analysis revealed a limited number of significantly altered genes in each cell line: 18 in HT-29 and 4 in MDST8 (**Table 1**). *DUSP1* emerged as the only gene consistently downregulated in both cell lines. Despite the limited DE response, gene set enrichment analysis (GSEA) identified divergent molecular programs activated by *Fn* in each subtype. HT-29 cells demonstrated upregulation of pathways related to ribosomal function, mitochondrial translation, and protein synthesis (**Figure 9A**), whereas MDST8 cells displayed enrichment of transcription factor and microRNA target gene sets (**Figure 9B**). These results suggest that *Fnn* employs distinct molecular mechanisms according to host tumour subtype. The consistent downregulation of *DUSP1* is notable given its role as a MAPK phosphatase that regulates inflammation, autophagy, and cell survival pathways (35,36). Together, these findings provide evidence for subtype-specific transcriptional responses, alongside a potential shared host stress response mechanism between CMS subtypes in *Fn*-positive CRC, contributing to our understanding of microbe-host interactions in colorectal cancer pathogenesis and therapeutic response.

Fnn has been increasingly recognised for its tumour-promoting effects in CRC, including modulation of host gene expression, induction of pro-inflammatory pathways, and promotion of epithelial to mesenchymal transition (EMT) (37–39). The differential responses observed between HT-29 and MDST8 cells are consistent with the known biological features of CMS1 and CMS4 subtypes. CMS1 tumours, such as those modelled by HT-29 (alternatively HCT116, LoVo (21,40)), are characterised by hypermutation, microsatellite instability and immune reactivity, showing a tendency toward adaptive immune response activation (4,20). The enrichment of ribosomal and translational gene sets in HT-29 may reflect activation of biosynthetic and immune-related pathways in response to infection, aligning with evidence that *Fn* promotes pro-inflammatory gene expression via NF- κB and MAPK signalling in epithelial cells (12,41). Additionally, the upregulation of *CYP1A1* and *AHRR* in HT-29 suggests activation of the aryl hydrocarbon receptor (AhR) signalling pathway (42,43). A recent CRC study demonstrated that *Fn*-derived formate activates AhR in HT-29 cells, inducing nuclear translocation and promoting cancer stem cell (CSC)-like properties through downstream Wnt/β-catenin activation (43). The observed upregulation of *ALDH1A3* in HT-29 further supports this mechanism. Elevated *ALDH1A3* expression and ALDH enzymatic activity were previously reported in *Fn*-treated HT-29 cells and colonic organoids, where they were mechanistically linked to formate-driven CSC phenotypes mediated via AhR and Wnt signalling (43). STRING network analysis in HT-29 cells showed strong functional connectivity between *ALDH1A3, CYP1A1* and *AHRR*, reinforcing the coordinated activation of the AhR signalling axis (**Figure 8A**).

In contrast, CMS4 tumours, modelled by MDST8 are defined by stromal infiltration, EMT and TGF- β signalling, as well by their immunosuppressive microenvironment,

typically driven by innate immune activation (4,20)**.** The observed downregulation of immediate early genes (IEGs) such as *EGR1* and *FOSB* (44) in MDST8 may reflect an alteration of acute transcriptional responses to *Fnn*. This is consistent with evidence that microbial stimuli, including viral infection, can lead to downregulation of these genes in a time- and context-dependent manner (45). STRING analysis of MDST8 also revealed a significantly enriched protein-protein interaction (PPI) network among *EGR1*, *FOSB* and *DUSP1* (**Figure 8B**), supporting a coordinated disruption of early transcriptional and MAPK-associated signalling in MDST8 (35). This was further reflected in GSEA results, showing significant negative enrichment of transcription factor targets regulated by *TFAM* and *GTF3A* (**Figure 9B**). However, as neither transcription factor was found to be differentially expressed, these findings likely reflect downstream effects and should be interpreted cautiously. The coordinated suppression of *TFAM* and *GTF3A* target genes may indicate a shift toward metabolically repressed and autophagic states in CMS4, aligning with their mesenchymal and treatment-resistant phenotype (46–49).

The consistent downregulation of *DUSP1* across both cell lines may indicate a common *Fnn*-induced mechanism impacting MAPK regulation. *DUSP1* is a dual specificity phosphatase that deactivates p38, JNK, and ERK MAPKs, and is a key regulator of inflammation, autophagy and epithelial stress responses in the gut (35,36). In CRC, *DUSP1* loss has been linked to enhanced proliferation and tumour growth in advanced stages, largely via sustained ERK/MAPK signalling (50). This microbial suppression of *DUSP1* may therefore enable prolonged MAPK activity, promoting a tumour-permissive environment across CRC subtypes. Notably, *Fn* has been shown to drive chemoresistance by inducing autophagy and suppressing apoptosis through TLR4/MYD88-dependent signalling, involving the modulation of miRNAs such as *miR-4802* and *miR-18a* (3,9). Given *DUSP1*'s regulatory role in autophagy pathways, its downregulation may contribute to this established pro-survival response, aligning with *Fnn*'s capacity to promote immune evasion and therapy resistance. Furthermore, *DUSP1* regulation is highly context-dependent, influenced by transcription factors such as NF- κB and AP-1, as well as by microRNAs and redox-sensitive modifications (35). Together, these findings suggest *DUSP1* may act as a microbial stress-responsive hub that *Fnn* exploits to modulate host cell fate and therapeutic vulnerability.

A key strength of this study lies in the approach of examining representative cell lines for distinct CMS subtypes, enabling exploration into how *Fn* may differentially modulate host transcriptional responses across CRC contexts. The use of established CRC cell lines representative of CMS1 and CMS4 subtypes offers a biologically relevant model to explore context-dependent host microbe interactions, an area previously underexplored in *Fnn* research (17,51). By integrating differential expression and gene set enrichment analyses (GSEA), the study reveals both subtype-specific and shared transcriptional programs that may influence tumour behaviour and treatment outcomes.

However, certain limitations must be acknowledged. The number of differentially expressed genes identified was relatively low; particularly in MDST8 cells. This may reflect intrinsic biological differences, such as a reduced transcriptional response to *Fnn* in CMS4-like tumours, as well as the influence of technical variability between samples. Principal component analysis (PCA) showed notable dispersion among

replicates in both cell lines, suggesting technical noise that may have limited statistical power for differential expression analysis **(Figure 6)**. Nevertheless, the higher number of differentially expressed genes observed in HT-29 cells, despite comparable variability, supports the interpretation that CMS1-like cells are more transcriptionally responsive to *Fnn* exposure under the conditions tested. This is consistent with the known biology of CMS1 tumours, which are typically hypermutated due to microsatellite instability and characterised by heightened immune activation and responsiveness to external stimuli (4). Additionally, as RNA-seq was conducted at a single post-infection timepoint (4 hours post-infection), only early transcriptional changes may have been captured, and later or sustained dynamic responses could have been missed. Finally, while HT-29 and MDST8 cell lines are established models for CMS1 and CMS4 subtypes respectively, they do not fully reflect the inherent complexity and heterogeneity of the in vivo CRC tumour microenvironment.

Despite these limitations, the study provides meaningful insights into how *Fn* may differentially interact with CRC subtypes at the transcriptional level. Previous studies have demonstrated that *Fnn* promotes inflammation, epithelial-to-mesenchymal transition (EMT) and chemoresistance in CRC by modulating host signalling pathways such as NF- κB, Wnt/ β-catenin and TLR4/MYD88, yet these findings have largely emerged from single CRC models without acknowledgement of molecular subtype stratification (9,12,14,15,37,38,52). In this study, *Fnn* exposure in HT-29 cells, representing the CMS1 subtype, led to enrichment of ribosomal and translational gene sets, along with activation of aryl hydrocarbon receptor (AhR) signalling. Notably , the upregulation of canonical AhR targets including *CYP1A1, AHRR* and *ALDH1A3* supports a role for this pathway in mediating *Fnn*-induced responses. *ALDH1A3* has been linked to aldehyde metabolism and cancer stemness, and its induction may further reflect a pro-tumorigenic shift (53). In contrast, MDST8 cells modelling CMS4 exhibited repression of immediate early genes and negative enrichment of transcription factor targets related to biosynthetic processes, suggesting a shift toward a low-proliferative, stress-adapted phenotype. Exploratory KEGG analysis, while not independently conclusive, supported these interpretations (**Figure 10**). HT-29 demonstrated enrichment of immune, stress-related and inflammatory pathways, whereas MDST8 showed enrichment in pathways associated with intercellular communication, neuroendocrine signalling and cellular plasticity; features that align with immune reactivity in CMS1 and the dynamic, migratory phenotype characteristic of mesenchymal CMS4 tumours. These divergent responses highlight the influence of tumour molecular context in shaping the host transcriptional response to *Fnn* and underscore the value of subtype-specific approaches when investigating host-microbe interactions in CRC.

Conclusion

This study reveals that *Fusobacterium nucleatum* subsp*. nucleatum* elicits distinct transcriptional responses in CRC subtypes, while also converging on a shared stress-related pathway involving *DUSP1* suppression. By stratifying cell line responses according to CMS subtype, this work addresses a notable gap in the literature, where *Fnn'*s effects have largely been studied in generalised or single-line CRC models. These findings highlight the importance of tumour context in shaping microbial influence on host gene expression, immune signalling and therapy resistance. While limited by sample size, technical variability, in vitro conditions and a single post-

infection timepoint, this study reinforces the value of integrating subtype-specific frameworks into cancer research. Future studies should validate these responses in primary tissues or organoid systems and explore whether microbiome-informed strategies, including bacterial targeting or modulation, could complement subtype-specific treatments. Together, these findings advance a refined understanding of how microbial influences, such as those of *Fnn,* contribute to tumour progression.

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